

HONG KONG BAPTIST UNIVERSITY

COURSE OUTLINE

1. COURSE TITLE

Innovative Laboratory

2. COURSE CODE

COMP7065

3. NO. OF UNITS

3 Units

4. OFFERING DEPARTMENT

Master of Science in Data Analytics and Artificial Intelligence

5. PREREQUISITES

COMP7035 Python for Data Analytics and Artificial Intelligence

COMP7015 Artificial Intelligence

COMP7990 Principles and Practices of Data Analytics

6. MEDIUM OF INSTRUCTION

E

7. AIMS & OBJECTIVES

This course covers laboratory works and mini-projects on data analytics and artificial intelligence. It provides students hands-on experience by applying programming skills and software tools to solve problems in data analytics and artificial intelligence.

8. COURSE CONTENT

I. Background topics in data analytics and artificial intelligence for the laboratory exercises and problem-solving exercises

- Data management and preprocessing: database access, data scraping, data preprocessing
- Quantitative analytics: Python Pandas and NumPy, statistical data analytics
- Data mining: association rule mining, similarity matching
- Artificial intelligence and machine learning: regression, classification and clustering, deep learning, generative AI
- Data visualization: Matplotlib, visualizing Images and 3D shapes, visualizing graphs

and networks

9. COURSE INTENDED LEARNING OUTCOMES (CILOs)

CILO	By the end of the course, students should be able to:
CILO 1	Characterize the process and importance of each topic in data analytics and artificial intelligence.
CILO 2	Analyze a real-world data-related problem and provide solutions to handle it.
CILO 3	Apply programming skills and software tools to solve some classic problems in data analytics and artificial intelligence.

10. TEACHING & LEARNING ACTIVITIES (TLAs)

CILO alignment	Type of TLA
1	Short briefing lectures. These lectures cover the basic concepts of various data analytics and AI tools.
1,3	Discovery laboratory exercises. Students will familiarize themselves with the basics and syntax of various tools, plug-ins, and libraries for various topics in data analytics and AI.
2,3	Problem-solving exercises. Students will apply a mixture of multiple tools, plug-ins, and libraries to solve some real-world problems.

11. ASSESSMENT METHODS (AMs)

Type of Assessment Methods	Weighting	CILOs to be addressed	Description of Assessment Tasks
Laboratory Reports	40 %	1-3	Reflection questions of laboratory reports are designed to assess whether a student is familiar with various tools, plug-ins, and libraries.
Mini-projects	60 %	1-3	Students are expected to form groups to devise a solution for a real-world problem with real-world data. It assesses how well a student can analyze a real-world problem and how well can he/she apply the techniques to devise a solution.

Last Update: 2024-06-26
Published Date: 2025-11-05

*** END ***